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SCI 171L: Science, Environment, and Society & BIO 211L: Evolution & Ecology: Species: Birds

This guide is for the creation of an ongoing student project chronicling native and non-native species found in northwest Indiana.

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Raptors (Owls, Hawks, Eagles, Falcons, Vultures)

Bald Eagle by Santiago Morales



Common Name: Bald Eagle

Scientific Name: *Haliaeetus leucocephalus*

The Bald Eagle is the national bird of the United States and is easily recognizable by its white head and tail contrasting with a dark brown body and wings.

Native: The Bald Eagle is native to North America, primarily found in Canada and the United States. They inhabit regions close to water, such as lakes, rivers, and marshes.

Conservation Status: Least Concern (IUCN Red List)

The Bald Eagle was once listed as Endangered in the lower 48 states, mainly due to hunting, habitat destruction, and the effects of pesticides like DDT. However, conservation efforts since the 1970s have helped bring their populations back, and they were officially removed from the endangered species list in 2007.

Ecological Role:

Trophic Level: The Bald Eagle is a top predator, which places it at the apex of the food chain. It primarily feeds on fish but will also scavenge carrion or prey on smaller birds and mammals.

Food Web: Bald Eagles help regulate populations of fish and smaller animals, contributing to the balance of aquatic ecosystems. They are also important scavengers, helping to clean up the environment by consuming carrion.

Special Role: As a scavenger, the Bald Eagle cleans up dead animals from its environment, contributing to the overall health of its ecosystem by preventing the spread of disease.

Bald Eagles are considered a keystone species in some regions, meaning they have a disproportionate impact on their environment compared to their population size.

Major Threats:

Habitat Loss: Deforestation, coastal development, and human encroachment on their habitats remain significant threats.

Pollution: Water pollution, especially from industrial runoff, can contaminate fish populations and poison the Eagles.

Climate Change: Shifting weather patterns and habitat disruption due to climate change can affect food availability and nesting sites.

While their populations have rebounded, these ongoing threats mean that the Bald Eagle remains vulnerable in some areas.

Fun Fact 1: Bald Eagles can live up to 30 years in the wild, and even longer in captivity.

Fun Fact 2: Despite their name, Bald Eagles are not "bald." The name refers to their white head (from the Old English word "balde," meaning white), not the lack of feathers.

Final Thoughts:

The Bald Eagle's recovery from near extinction is a testament to the power of dedicated conservation efforts. While their numbers have greatly increased, ongoing habitat protection and environmental conservation are crucial to ensuring the survival of this iconic bird. As a keystone species, the Bald Eagle plays a vital role in maintaining the health and balance of its ecosystem.

Red-Tailed Hawk by Anthony Hughes



Common Name: Red-Tailed Hawk

Scientific Name: *Buteo jamaicensis*

Native Status: The Red-tailed Hawk is native (indigenous) to North America. It is one of the most widespread and adaptable hawk species in the Americas, ranging from Alaska and Canada through the United States and into Central America and parts of the Caribbean. It is considered cosmopolitan within North America because it thrives in many habitats, including forests, grasslands, deserts, farmland, and even urban areas.

Conservation Status: The Red-tailed Hawk is listed as Least Concern. Its population is stable and relatively large due to its adaptability and broad diet. Unlike many raptor species, it has adjusted well to human-altered environments.

Role in the Ecosystem: Trophic Level: Tertiary Consumer (Carnivore / Apex or Near-Apex Predator)

Red-tailed Hawks primarily eat:

- Rodents (mice, voles, squirrels)
- Rabbits
- Snakes
- Small birds

Ecological Importance: The Red-tailed Hawk plays a crucial role in maintaining ecosystem balance:

- Population Control: Helps regulate rodent populations, which protects crops and limits disease spread.
- Indicator Species: Healthy hawk populations often indicate a healthy ecosystem.
- Top Predator Role: As a higher-level predator, it helps stabilize food webs.

Because it sits near the top of the food chain, toxins in the environment can accumulate in its body (a process called biomagnification).

Major Threats: Although currently stable, Red-tailed Hawks face several potential threats:

1. Habitat Loss--Urban expansion and deforestation.
2. Vehicle Collisions--Often hunt along highways.
3. Rodenticide Poisoning--Secondary poisoning from eating rodents exposed to poison.
4. Climate Change--Shifting prey availability and habitat conditions.

Historically, raptors suffered during the mid-20th century due to pesticides like DDT, but bans on harmful chemicals helped populations recover.

Interesting Facts:

- The famous "hawk scream" used in many movies is actually the call of the Red-tailed Hawk--even when other birds like eagles are shown on screen.
- Juvenile hawks do not have the distinctive red tail; it develops as they mature.
- They have extremely sharp vision--about 8 times better than human eyesight.
- Wingspan can reach up to 4.5 feet (1.4 meters).

Why This Species Matters: The Red-tailed Hawk demonstrates how wildlife can adapt to human development when ecosystems remain partially intact. Its role as a predator makes it vital for controlling prey populations and maintaining ecological balance. Protecting this species ensures continued health of grassland, forest, and even suburban ecosystems.

Corvids (Crow Family)

Blue Jay by Aaliyah Keil

Common name: Bluejay

Scientific Name: *Cyanocitta cristata*

The bluejay is native to North America. It is a commonly seen species with a stable population.

Trophic Level: Omnivore, Secondary Consumer

Bluejays are important seed dispersers, since they store acorns and then sometimes forget where they store them, allowing trees to grow. They are known as opportunistic feeders because they eat insects, nuts, fruits, and eggs of other birds. They act as alarm callers by warning other birds of nearby predators.

Major Threats: There are no immediate threats towards bluejays, however, they do face habitat loss, window collisions, and West Nile virus.

Interesting Facts: Bluejays are highly intelligent and capable of mimicking the calls of hawks to scare away other birds from food sources. They have a strong family bond and often work together to protect their nests. Their blue coloration is not due to pigment but rather the way light interacts with the microscopic structure of their feathers.

Ardeidae (Herons)

Common Name: Great Blue Heron

Scientific Name: *Ardea Herodias*

The great blue heron is native and Indigenous to North America as well as Central America. It is considered cosmopolitan. This animal is found from Alaska and Canada all the way through the Caribbean. It stretches even as far south as the northern part of South America. The population of Great Blue Heron is stable and is known to be slightly increasing in many areas.

The ecological role of the Great Blue heron is that of a top predator in its ecosystem. It is an indicator species in aquatic ecosystems. As a predator(carnivore) in these ecosystems, it helps control the population of fish, amphibians and small mammals. It has a secondary trophic level. Some special aspects of blue heron is that they hunt by slowly wading or standing still in shallow water before striking with their beak.

Two of the more prominent threats to these animals is habitat loss and pollution. Drainage and shoreline development reduces the opportunity for nesting and feeding grounds for these beautiful birds. The final threat to these animals would be the threat that is presented to the whole environment, which is climate change. Climate change can alter water levels and prey distribution for the Blue Heron.

Interesting Facts:

- The Great Blue Heron has a wingspan of up to 6.5 feet but only weighs 5-6 pounds
- They can swallow fish longer than their own neck by flipping them and swallowing the prey headfirst
- They can use their beak as a spear when they feel threatened

Galliformes (Game Birds)

Bobwhite Quail by Victor Rodriguez

Common Name: Bobwhite Quail

Scientific Name: *Colinus virginianus*

They are native to Canada, Cuba, Guatemala, Mexico, and the US. Bobwhite quails are a near threatened species as listed by The IUCN Red List of Threatened Species in 2021. They are a prey species that control insect populations, disperse seeds, and serve as a food source for hawks, foxes, and snakes. They are primary or secondary consumers (omnivores). Northern Bobwhites' main threats are habitat loss, invasive species like fire ants, and climate change. Some fun facts are that they have an iconic Bobwhite call that's produced by the males to attract mates and mark territory. There are also 22 recognized subspecies, and among the most intensely studied birds in the world.

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